Using Stimulus to Manage Forms

Turbo is designed to allow regular form submissions to trigger interactive changes on a page without having to reload the entire page. Sometimes, though, you need a little more client-side zest. You might want the form to be submitted on a user action other than clicking a Submit button. Or you might want to gather data from elsewhere on the page. In either case, you can use Stimulus to mediate your form submission.

If you look at the schedule page of our application, it has a search bar that currently does nothing except maybe offer typing practice. What we'd like it to do is return search results on typing. Specifically we want the following functionality:

- Typing in the box triggers a form submission with search results.
- Receiving search results uses Turbo Frames to place those results in a modal that overlays our page.
- Clearing the search field clears the modal.
- Clicking outside the modal window clears the modal.

For those of you who use the Hey email app,[63] this functionality is based on Hey's search functionality, though this is a simplified version of it.

To make this work, we need a polyfill. Generically, a *polyfill* is code that allows you to use a feature even if that feature has not been implemented on all browsers. Typically, the polyfill checks to see if the feature exists, and if it doesn't, the polyfill implements the feature (or a stripped-down version of it) on the browser so that code can be more easily shared between different browsers.

There's a slight difference in browser APIs as I write this, in that Apple's Safari browser does not implement a method we need to handle form submissions correctly. To rectify that, we need to use the polyfill package.